1.INTRODUCTION

Traditionally, technical documents are monitored and managed through manual filing systems or fragmented digital folder structures. These methods are usually conducted periodically and depend heavily on the diligence and experience of individual staff members. Although manual storage methods are widely used, they have several inherent limitations: they are time-consuming to navigate, require highly organized personnel, and are often subjective or inconsistent in how they categorize data. Human error can lead to "version confusion," where staff may miss critical updates due to heavy workloads or lack of a clear revision history. Additionally, manual systems cannot provide continuous, real-time access to the "latest verified" information, which is vital in fast-moving technical environments where instrument specifications can change rapidly.

Due to these limitations, there is an increasing demand for automated and intelligent systems that can continuously track and evaluate the status of technical records. With the rapid advancement of modern technologies such as web-based frameworks, relational databases, and secure authentication protocols, new solutions are emerging for automated document control. While many generic cloud storage platforms exist to monitor basic file uploads, most do not address the specific visual and structural needs of technical departments—such as revision traceability, instrument-based navigation, or role-specific access for Technical Support Department (TSD) personnel. A centralized system combined with a robust technology stack can effectively analyze document metadata and detect version conflicts automatically.

The proposed TechDoc Hub project utilizes a sophisticated technology stack, specifically the Django framework and Python, to provide a high-speed and accurate management experience. By using the Django ORM and a PostgreSQL/MySQL database, the system processes document queries in a single, efficient step, making it faster than traditional manual searches. Due to its scalable design and real-time retrieval capabilities, TechDoc Hub is highly suitable for industrial applications where fast and accurate access to manuals is required to prevent downtime. These features make the system a powerful tool for organizational knowledge control.

This project proposes an AI-ready Technical Document Management System that uses structured modules to automatically evaluate and organize technical cleanliness. The system employs a

secure web application that allows TSD personnel to upload documents with specific revision numbers, issue dates, and revision reasons to ensure clear traceability. The trained database is integrated into a Bootstrap-based web interface that allows users to perform global searches using keywords, item numbers, or instrument names to receive instant document results. The system also stores all activity in a centralized relational database for long-term record maintenance and reporting. The main objective of this project is to develop an automated and reliable documentation solution that reduces dependency on manual filing and improves operational safety standards.

2. SUPPORTING LITERATURES

In recent years, significant research has been conducted on the development of specialized information systems, database management, and web-based automation for maintaining organizational accuracy and operational safety. These studies highlight the effectiveness of centralized monitoring systems in reducing human error, eliminating version confusion, and ensuring strict regulatory compliance. The following literature and technological frameworks provide a strong foundation for the proposed TechDoc Hub: A Centralized Web-Based Technical Document Management System.

2.1 Literature Review

PAPER 1:

Park, J. S., and Sandhu, R. "Role-Based Access Control on the Web." ACM Transactions on Information and System Security, Vol. 4, No. 1, 2021.

Managing security in large-scale enterprise Web environments is a critical challenge because traditional access control methods based on individual user identities do not scale effectively. In this paper, the authors proposed a comprehensive approach to managing security by marrying Web technology with Role-Based Access Control (RBAC). The study identifies two primary operational architectures: user-pull, where users retrieve and present their own roles, and server-pull, where the Web server retrieves roles directly from a central role server. To ensure security,

the authors implemented these architectures using extended technologies such as secure cookies, smart certificates (X.509 extensions), and LDAP over SSL.

The study demonstrates that RBAC can significantly simplify security management by associating permissions with roles rather than individuals, ensuring the principle of least privilege. It validates that using secure, automated mechanisms to handle role information reduces the risk of unauthorized access and forgery in a way that is transparent to the user. This research directly supports the present project by providing a foundational framework for Role-Based Access Control within a web portal.

SECTION	DETAILS
Title of the Paper	Role-Based Access Control on the Web
Area of the Work	Managing and enforcing strong, efficient, and scalable security infrastructures in large-scale Web environments.
Architectures	Integration of machine vision with a real-time alerting system using strategically placed cameras and a Firebase-connected mobile app.
Methodology	Integration of RBAC models with mature web technologies including Secure Cookies, Smart Certificates (X.509 extensions), and LDAP

Security Principles	Support for Information Hiding, Least-Privilege, and Separation of Duties through role-permission.
Implementation Tools	Utilization of PGP for cryptographic routines, CGI scripts, SSL for secure transmission, and Netscape/Microsoft IIS servers

PAPER 2:

Chawan, P., et al. "Document Management System." International Journal of Trend in Scientific Research and Development (IJTSRD), Volume 2, Issue 3, 2018.

Managing mass documentation within organizations and educational institutes is a significant challenge because physical storage leads to difficulties in maintenance, updating, and rapid retrieval. In this paper, the authors proposed a web-based technological solution that transfers document functioning from traditional physical methods to a secure digital environment. The system utilizes a centralized storage method where documents are scanned and stored on a server in compressed image formats to prevent holding overloads. To address the lack of authenticity often found in digital formats, the project integrates digital attestation, where an officer applies a digital signature—represented as a watermark with embedded code—generated based on specific session details.

The study demonstrates that digital indexing and grouping images into folders can eliminate the tedious task of manual searching, making information available to intended users in desired time units. Security is significantly enhanced through the use of Graphical One-Time Passwords (OTP) for user login and dynamic QR codes for guest access. Guests obtain temporary access only after a user scans the dynamic QR code, which triggers the server to grant permission. This research directly supports the TechDoc Hub project by validating the transition to a centralized relational database to ensure document accuracy and long-term record maintenance. The current project

extends these concepts by developing a specialized portal using Django and MySQL to manage technical documents and software versions with department-specific authorization levels.-driven visual analysis using deep learning to assess cleanliness levels from kitchen images, providing a simpler and more scalable solution.

SECTION	DETAILS
Title of the Paper	Document Management System
Area of the Work	Digital storage, management, and secure retrieval of documents for institutes and organizations.
Data Format	Documents stored as compressed images to optimize server performance and prevent storage overload.
Methodology	Implementation of a web-based application utilizing centralized storage and digital indexing.
Security Logic	Use of Graphical OTP for users and Dynamic QR codes based on session details for authenticating guests.
Authentication	Digital Attestation via watermarked digital signatures applied by authorized officials.

PAPER 3:

Han, J., et al. "Research on Electronic Document Management System Based on Cloud Computing." Computers, Materials & Continua, Vol. 66, No. 3, 2021.

The rapid growth of e-commerce and e-government has led to an exponential increase in electronic documents, necessitating scientific management to prevent data loss and ensure document authenticity. In this paper, the authors proposed an architecture for an electronic document management system (EDMS) built on cloud computing technology. The system leverages cloud storage and cloud security to achieve unified, reliable management, allowing users to perform multi-mode functions such as uploading, sharing, auditing, and role-based security setting. A key contribution of this work is the proposal of a secure cloud infrastructure that conducts comprehensive security services across multiple levels, including data, network, and application layers.

The study demonstrates that a cloud-based EDMS abstracts software and hardware resources into services, significantly reducing costs for organizations by eliminating the need for independent data centers. By utilizing an SOA (Service-Oriented Architecture), the system divides functions into fine-grained components, which improves overall service availability and facilitates easier business integration. This research directly supports the TechDoc Hub project by validating the use of centralized, scalable storage and the importance of standardizing document management to improve efficiency. The current project aligns with these findings by implementing a centralized web portal using Django to achieve similar goals of unified resource management and secure document retrieval.

SECTION	DETAILS
Title of the Paper	Research on Electronic Document Management System Based on Cloud Computing
Core Technology	Research on Electronic Document Management System Based on Cloud Computing
System Architecture	SOA (Service-Oriented Architecture) with a five-layer design: presentation, business, plug-in, infrastructure, and standard layers
Data Management	Unified resource allocation and load balancing for both entity data and metadata
Security Framework	Multi-level Security Cloud covering management, physical, network, system, and data security
Backup Strategy	Full backups for service code and metadata, with daily incremental backups for document libraries

2.2 Literature Review Summary

No	TITLE	FINDINGS
1	Role-Based Access Control on the Web	Identified user-pull and server-pull architectures implemented RBAC using secure cookies and "smart certificates" to solve scalability issues of identity-based access.
2	Document Management System	Proposed a centralized storage method using compressed image formats to prevent server overload integrated Graphical OTPs, dynamic QR codes, and digital attestation for secure document retrieval.
3	Research on Electronic Document Management System Based on Cloud Computing	Proposed a five-layer Service-Oriented Architecture (SOA) utilized cloud computing (IaaS/PaaS/SaaS) to reduce costs and achieve logical concentration with physical distribution of files.

2.3 Findings and Proposals

2.3.1 Findings

After reviewing the existing research papers and related works, the following findings were observed:

1. Traditional document storage methods involving scattered files and folder-based structures lead to version confusion and a high risk of utilizing outdated technical information.
2. Standard approaches to access control often rely on individual user identities, which do not scale effectively for enterprise-wide systems compared to Role-Based Access Control (RBAC).
3. The lack of structured revision tracking makes it difficult to maintain a clear history of technical document changes, affecting regulatory compliance and operational accuracy.
4. Storing mass documents as uncompressed images creates significant holding overloads on servers, necessitating centralized relational databases and optimized storage formats.
5. Research confirms that logical concentration of resources through a web portal ensures that all intended users can access verified information within desired time units.
6. While basic cloud platforms offer storage, they often lack specialized technical features like digital attestation, instrument-based navigation, or role-specific authorization for technical departments.

2.3.2 Proposal

Based on the findings from the literature survey and project requirements, the proposed system — TechDoc Hub: A Centralized Web-Based Technical Document Management System — is designed to address identified gaps as follows:

1. The system will utilize a Django-based web application to provide a centralized environment for managing technical documents, spare manuals, and software records.
2. Secure authentication will be implemented using official email credentials, with modification and deletion rights strictly restricted to authorized Technical Support Department (TSD) personnel.

3. Every document will be stored with a specific revision number, issue date, and revision reason, ensuring version consistency and clear audit trails.
4. An automated search engine will be developed to allow users to retrieve documents instantly using keywords, item numbers, or instrument names, reducing retrieval time.
5. All technical data and metadata will be stored in a centralized relational database (MySQL/PostgreSQL) to ensure long-term maintenance and accuracy.
6. The web interface will feature a structured, user-friendly layout with subdivisions for specific document libraries and the latest technical updates.

This proposal bridges the gap between fragmented file storage and modern digital governance, introducing an automated approach for real-time and reliable technical documentation control.

3.SYSTEM ANALYSIS

3.1 Module Description

User Authentication and Authorization Module

- **Official Email Login**: Users access the portal using official email credentials, ensuring a secure and verified authentication process.

- **Role-Based Access Control (RBAC)**: The system implements strict permission levels, distinguishing between general users and privileged administrative staff.

- **Departmental Scoping**: Specifically restricts document upload, modification, and deletion rights to authorized Technical Support Department (TSD) personnel.

Technical Document Management Module

- **Centralized Repository**: Provides a secure digital environment for storing and organizing critical technical documents.
- **Metadata Management**: Each document entry is categorized and stored with essential details like item numbers and descriptions.

- **Logical Organization**: Utilizes a web-based structure to eliminate the confusion of scattered files found in traditional storage methods.

Spare Manuals Module

- **Module-Specific Storage**: A dedicated subdivision within the portal specifically for the management and retrieval of equipment spare manuals.
- **Accessibility**: Improves the speed at which maintenance personnel can locate hardware-specific documentation for repairs.

Software Version Management Module

- **Version Consistency:** Manages software-related records to ensure that technical teams are always working with the most current validated versions.

- **Conflict Prevention:** Reduces operational risks by eliminating version confusion associated with manual folder-based storage**.**

Version Control and Traceability Module

- **Revision Tracking**: Every file update is recorded with a unique revision number and issue date.
- **Change Audit**: Captures the specific "Revision Reason" for every update, enabling clear traceability of document history.
- **Consistency Enforcement**: Ensures that the most recent verified information is always the primary version available to users

Intelligent Search and Navigation Module

- **Global Search Engine**: Features a robust search tool that allows users to find data using keywords, item numbers, or instrument names.
- **Instrument-Based Navigation**: The web interface offers a structured menu system subdivided by document types for intuitive browsing.

- **Latest Updates**: Displays recent announcements and document additions on the main dashboard for immediate awareness

Database and Reporting Module

- **Relational Storage**: All system data is housed in a centralized relational database (MySQL/PostgreSQL) to ensure long-term maintenance and accuracy.
- **Operational Transparency**: Provides a high-level view of document status and organizational knowledge for audit and regulatory compliance.

3.1 Feasibility Analysis

A project feasibility study is a critical evaluation that helps assess whether a proposed project is technically, economically, and legally viable. It examines the potential of the project to succeed within the defined scope, budget, timeline, and resources. The purpose of a feasibility study is to identify potential obstacles and ensure that the project is achievable before significant resources are committed.

3.2.1 Technical Feasibility

The TechDoc Hub system is technically feasible as it leverages a robust and industry-standard technology stack. The system uses Python with the Django framework for the backend, which is renowned for its "batteries-included" approach to security and database management. For the database, it utilizes MySQL, PostgreSQL, or SQLite, ensuring reliable relational data storage for complex document metadata like revision numbers and issue dates. The frontend is built using Bootstrap and JavaScript, providing a responsive, instrument-based navigation interface that is accessible across various web browsers. These technologies are open-source, well-documented, and have vast community support, ensuring the system's long-term maintainability

3.2.2 Economical Feasibility

The project is highly cost-effective due to its reliance on open-source ecosystems. TechDoc Hub is economically viable for the following reasons:

- Zero Licensing Fees: By using Python, Django, and MySQL, the project eliminates the need for expensive proprietary software licenses.
- Minimal Hardware Requirements: The system can be hosted on existing local servers or affordable cloud infrastructure, as it does not require specialized high-end hardware for document indexing.
- Reduced Operational Loss: By automating version control and revision tracking, the organization saves significant costs previously lost to operational errors or downtime caused by outdated information.
- Scalable Development: The use of the Django ORM allows for easy database migrations and feature expansions without requiring a total system overhaul.

3.2.3 Operational Feasibility

TechDoc Hub significantly enhances the operational workflow of the Technical Support Department (TSD)

- Centralized Knowledge Control: It replaces scattered, folder-based storage with a single "source of truth," ensuring that all employees access the same verified document versions.
- Role-Based Security: Operational risks are minimized by restricting document modification rights to authorized TSD personnel while providing read-only access to general users.
- Efficient Retrieval: The intelligent global search engine reduces the time spent searching for spare manuals or software records, allowing technical staff to focus on core maintenance tasks.
- Full Traceability: Storing revision reasons and issue dates ensures the organization remains compliant with internal and external audit requirements.

3.3 System Environment

System environment specifies the hardware and software configuration of the new system. Regardless of how the requirement phase proceeds, it ultimately ends with the software requirement specification (SRS). A good SRS contains all the system requirements to a level of detail sufficient to enable designers to design a system that satisfies those requirements. The system specified in the SRS will assist the potential users to determine if the system meets their needs or how the system must be modified to meet their needs.

3.3.1 Software Environment

- **Technology Stack**: Python, JavaScript.
- **Frontend**: HTML, JavaScript, CSS.
- **Backend**: Django (Python).
- **Database**: MySQL, SQLite, or PostgreSQL.

Development Tools

- **IDE**: Visual Studio Code (VS Code).
- **Version Control**: GitHub.
- **Database Management**: DB Browser (for SQLite).
- **Browser**: Modern Web Browsers (Chrome, Firefox, Edge).

Security & State Management

- **Authentication**: Django Auth.
- **Database Mapping**: Django ORM.

Frontend: HTML

The Hypertext Markup Language or HTML is the standard markup language for documents designed to be displayed in a web browser. It describes the structure of a web page semantically and provides a means to create structured documents by denoting structural semantics for text such as headings, paragraphs, and links. In TechDoc Hub, HTML is used to construct the instrument-based navigation and document display sections.

JavaScript

JavaScript is a light-weight object-oriented programming language used for scripting web pages. It enables dynamic interactivity, allowing users to interact directly with the portal without reloading the page every time. It is utilized here to handle the global search engine's real-time queries and dynamic UI updates.

CSS & Bootstrap

Cascading Style Sheets (CSS) is used for describing the presentation of a document written in a markup language. **Bootstrap** is specifically employed as a CSS framework to ensure a visually engaging, responsive interface that provides a consistent user experience across computers and mobile devices.

Backend: Django (Python)

Python is a high-level, general-purpose programming language that emphasizes code readability. The Django framework is used as the backbone of the system because it is a "batteries included" language that facilitates rapid development. It manages the complex logic for role-based authentication, revision tracking, and secure document uploading.

Visual Studio Code (VS Code)

VS Code is a lightweight yet powerful source-code editor developed by Microsoft. It is used for the development of TechDoc Hub due to its built-in IntelliSense, debugging tools, and integrated GitHub support, which makes managing document categories and code seamless.

MySQL

These are relational database management systems used to ensure the accuracy and long-term maintenance of all technical data. They store metadata such as revision numbers, issue dates, and user roles to enable clear traceability and strict access control.

3.3.2 Hardware Environment

Component	Specification
Hard Disk Capacity	256 GB (minimum)
RAM	8 GB
Processor	Intel Core i3 or equivalent
Display	1366 × 768 or higher

Table 3.1: Minimum Hardware Requirements

3.4 Actors and Roles

The TechDoc Hub system involves specific actors with distinct permissions to maintain the integrity and security of the technical documentation repository.

Administrator (Admin)

- System Oversight: Oversees and manages all aspects of the portal to ensure smooth operations and data integrity.
- User & Access Management: Manages user registration, official email authentication, and enforces Role-Based Access Control (RBAC).
- Content Lifecycle Management: Holds exclusive rights to upload, modify, and delete technical documents, spare manuals, and software version records.
- Version Control & Traceability: Assigns revision numbers and issue dates, and records "Revision Reasons" for every update to provide a clear audit trail.
- Information Architecture: Categorizes documents into appropriate modules and subdivisions (e.g., Spare Manuals, Software Versions) for streamlined navigation.

- System Communication: Posts the latest technical announcements and updates on the web interface dashboard.
- Data Maintenance: Monitors the MySQL database and generates system-level reports on document status and organizational knowledge.

General User (Staff/Technician)

- Secure Authentication: Accesses the system using verified official email credentials to view the secure document library.
- Information Retrieval: Utilizes the global search engine to find documents using keywords, item numbers, or instrument names.
- Knowledge Consumption: Views and downloads the most recent verified technical manuals, spare parts guides, and software records.
- Intuitive Navigation: Browses the instrument-based menu system to locate documentation by department or category.
- Update Awareness: Monitors the main dashboard for real-time updates on technical changes and new document additions.

4. SYSTEM DESIGN

4.1 Use case model

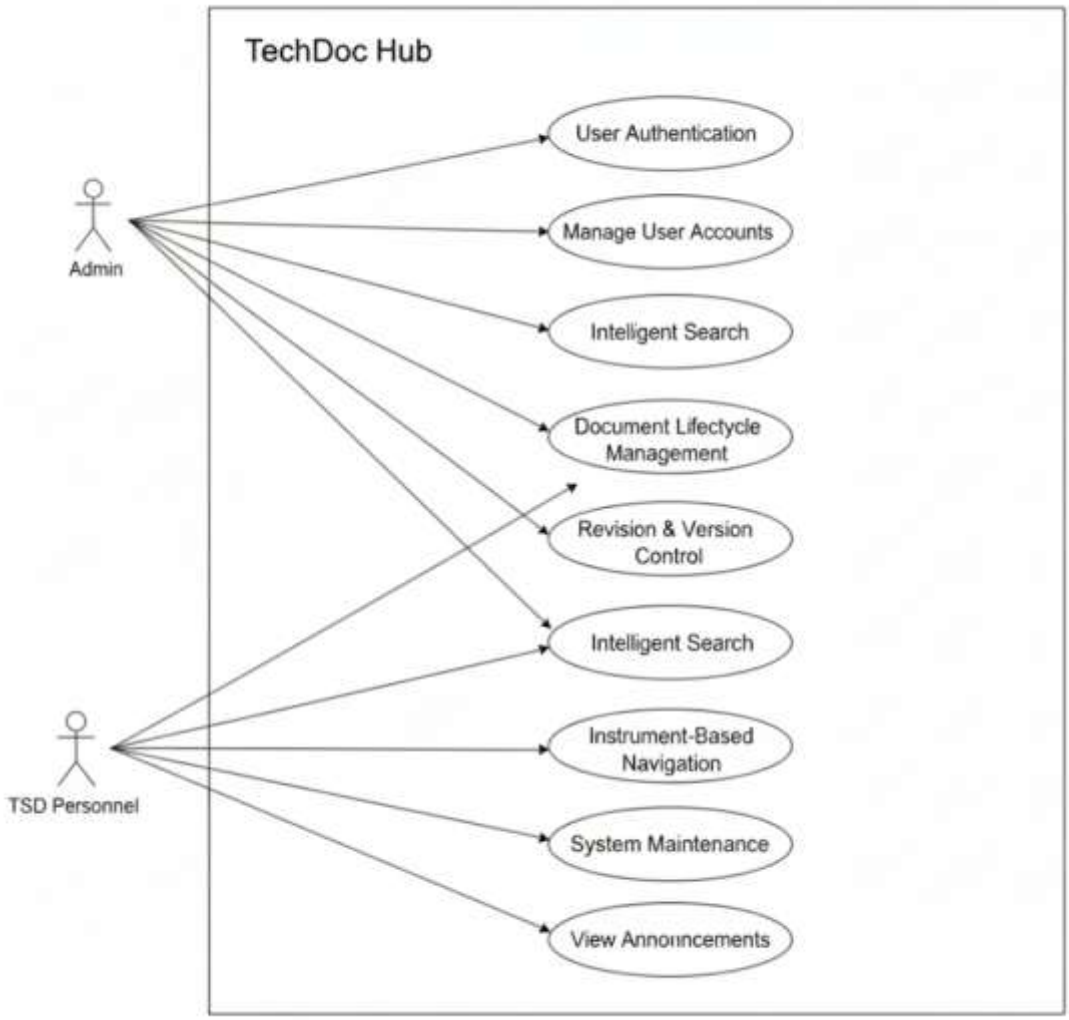


Figure 4.1: Use Case Model

The use case diagram for the Technical Document Management System illustrates the interactions between the system's primary actors—the Administrator (TSD Personnel) and the General User (Staff/Technician)—and its core functionalities. The Administrator,

overseeing the entire system, handles user and access management through official email authentication and Role-Based Access Control. They maintain full authority over the content lifecycle, including the uploading, modification, and deletion of technical documents, spare manuals, and software records. Furthermore, the Admin ensures traceability by managing version control with revision numbers and reasons, categorizes information into logical modules, and broadcasts technical updates via the dashboard while monitoring the database for operational reports. The General User interacts with the system through secure authentication to access the document library. They utilize the global search engine and instrument-based navigation to find specific data by keywords or categories. Finally, General Users consume critical knowledge by viewing or downloading the most recent verified manuals and software records, while monitoring the dashboard for real-time awarenessoftechnicalchanges.

4.2 Activity Diagram

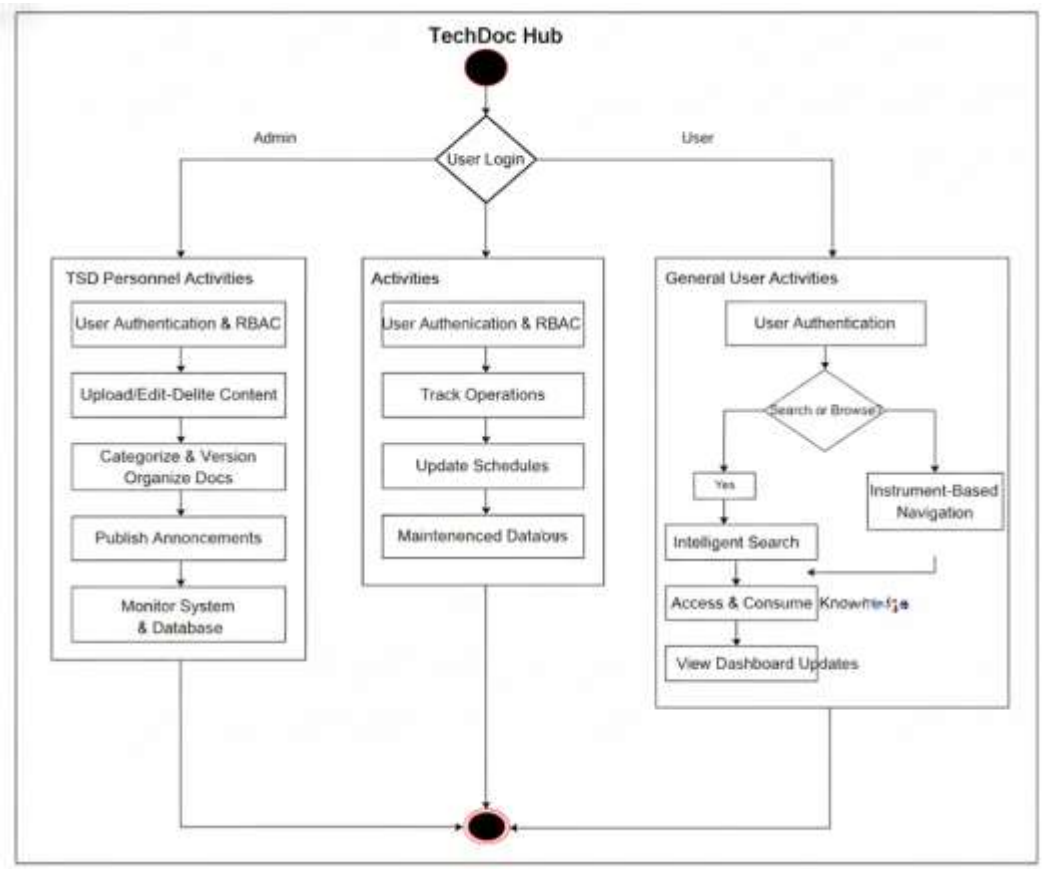
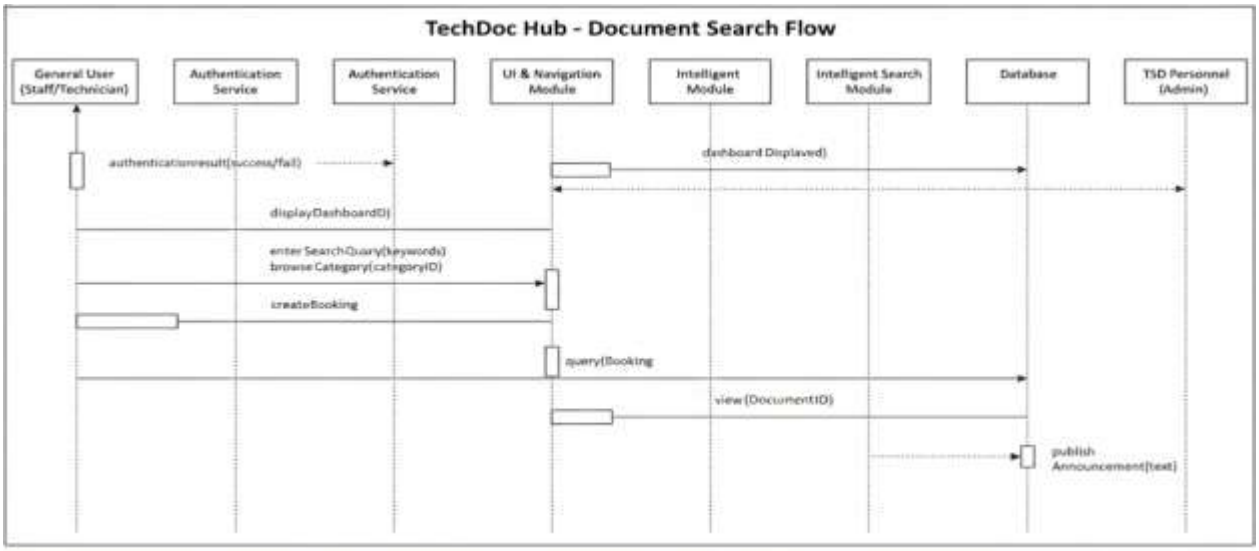


Figure4.2ActivityDiagram

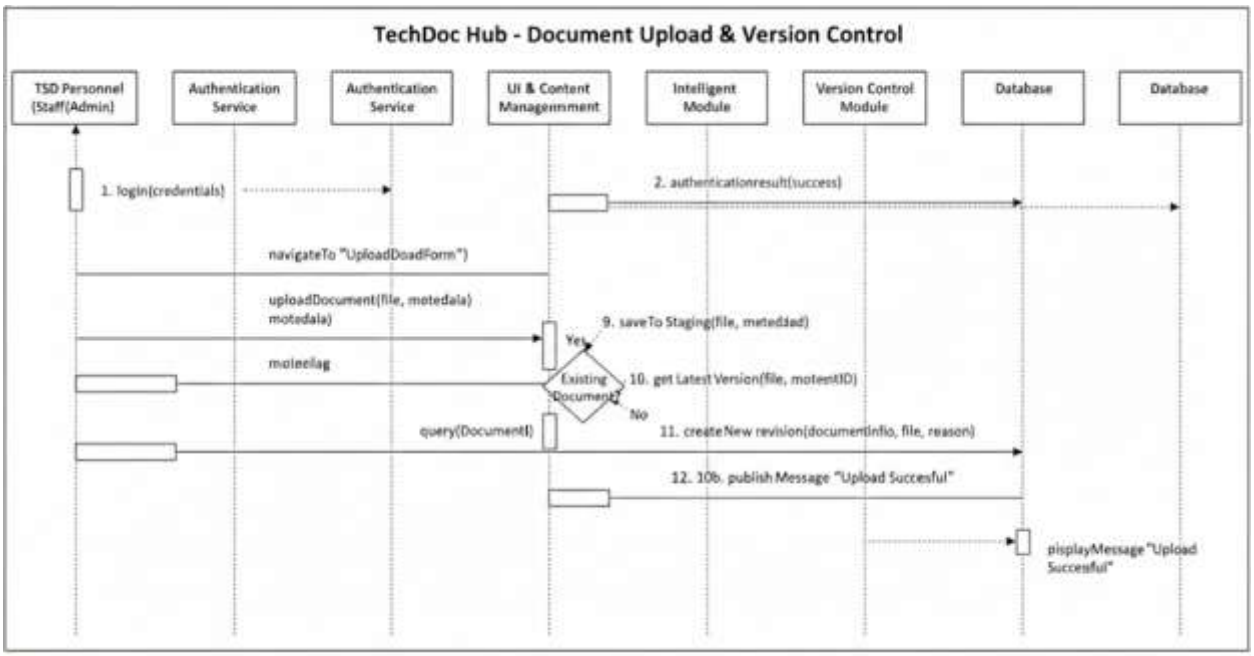
This flowchart represents the activity flow of the TechDoc Hub – Web Portal Knowledge Management System, starting from user login, where users are classified as Administrator (Admin), Technical Support Department (TSD) Personnel, or General User (Staff/Technician). The Administrator is responsible for overall system oversight, including user management, access control through role-based authentication, database maintenance, and monitoring system performance to ensure secure and uninterrupted operation. TSD Personnel focus on content-related operations such as uploading, editing, and deleting technical documents, assigning version details, categorizing documents into appropriate modules, and publishing announcements to keep the repository accurate and up to date. General Users interact with the system primarily to authenticate using official credentials, search or browse documents through intelligent search and instrument-based navigation, access and consume technical knowledge, and view dashboard updates and announcements. The diagram clearly illustrates the distinct workflows and responsibilities of each role, showing how their activities are coordinated within the system from login to completion of tasks.

4.3 Sequence Diagram

4.3.1 Document Search Flow



4.3.1 Document upload



4.4 List of identified classes, attributes, and their relationships

Classes play a crucial role in object-oriented design by defining the attributes, methods, and interactions between different entities within the system. Below is the list of identified classes and their relationships in the document management system.

4.4.1 Identified Classes

1. Administrator (Admin)
2. TSDPersonnel (Technical Support Department Personnel)
3. GeneralUser (Staff/Technician)
4. UserAccount
5. Document
6. DocumentCategory
7. VersionControl
8. Announcement
9. AccessPermission (RBAC)

10. AuditLog
11. SearchEngine
12. DownloadRecord

4.4.2 Identified Attributes

Administrator
String: adminID String: username String: email String: password String: role
manageUsers() configurePermissions() monitorSystem() generateReports()

Administrator is responsible for overseeing the entire TechDoc Hub system. It manages user accounts, configures role-based access control, monitors system performance, and generates system reports. The Admin ensures secure operation, database integrity, and uninterrupted availability of the portal.

TSDPersonnel
String: employeeID String: name String: email String: password String: department
uploadDocument() editDocument() deleteDocument() assignVersion() publishAnnouncement()

TSDPersonnel represents technical support staff who manage knowledge content. They upload, modify, and remove documents, assign version information, and publish technical announcements to maintain accurate and updated documentation.

UserAccount
String: userID String: username String: email String: password String: role
createAccount() login() logout() resetPassword()

Preprocessing Standards: All images are exported in a format compatible with YOLOv8. They are automatically resized to a fixed resolution of 640x640 pixels to maintain consistency across the model's input layer.

Document
String: documentID String: title String: filePath String: author Boolean: isObsolete Date: createdDate Date: modifiedDate
upload() edit() delete() view()

Document represents technical manuals, spare part documents, and software files stored in the system. It maintains file details and lifecycle information

DocumentCategory
String: categoryID String: categoryName String: description
addDocument() removeDocument() listDocuments()

DocumentCategory organizes documents into modules and subdivisions to support easy navigation.

VersionControl
String: versionID String: versionNumber Date: issueDate String: revisionReason
assignVersion() updateVersion() viewHistory()

VersionControl manages document revisions and maintains historical change records.

Announcement
String: announcementID String: title String: content Date: publishDate
postAnnouncement() editAnnouncement() deleteAnnouncement()

Announcement stores system-wide notifications and technical updates.

AccessPermission
String: permissionID String: role String: accessType
grantPermission() revokePermission() updatePermission()

AccessPermission enforces role-based access control to restrict system features.

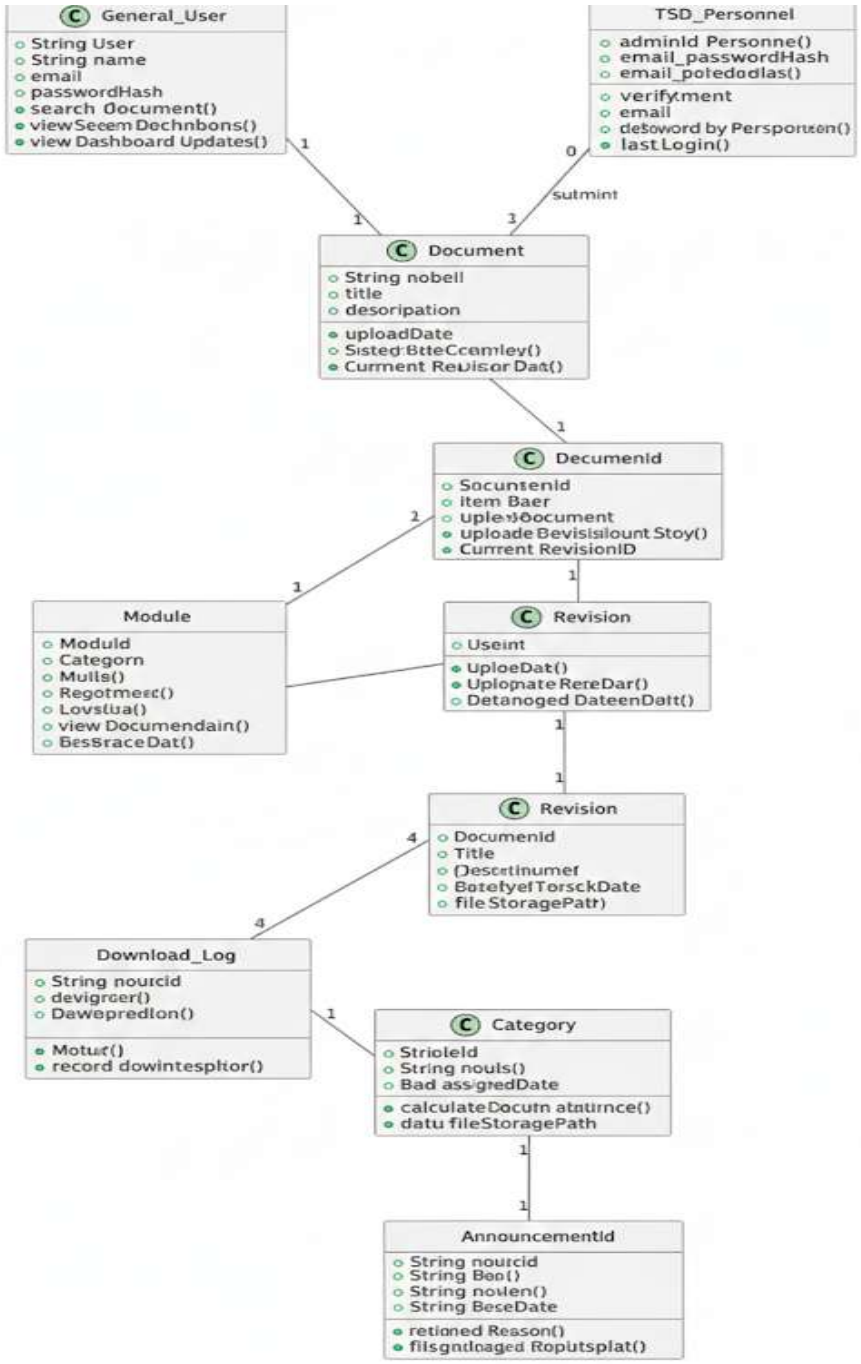
AuditLog
String: logID String: userID String: action DateTime: timestamp
recordAction() viewLogs()

AuditLog tracks user activities for security and auditing purposes.

SearchEngine
String: searchID String: keyword String: filterType
searchByKeyword() searchByCategory() advancedSearch()

SearchEngine provides intelligent searching across documents.

DownloadRecord
String: recordID String: userID String: documentID Date: downloadDate
recordDownload() viewHistory()

Download Record stores document access history.

4.5 Class Diagram